

HE2003 Econometrics I

Quiz Practice 1 (Solutions)

AY 2026/2027, Semester 1

Quantitative Research Society @NTU

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Overview

- **Coverage:** Lectures 1–5 and Lecture 6 slides 1–3 (large-sample distribution of $\hat{\beta}_1$ and its standard error); no t-tests, confidence intervals, p-values or Stata.
- **Format:** 5 questions, 50 marks total, approximately 60 minutes.
- **Standard normal table** ($\Phi(z) = P(Z \leq z)$ for $Z \sim N(0, 1)$), needed only where a probability is asked:

z	0.50	1.00	1.28	1.50	1.645	1.96	2.00	2.50	2.58	3.00
$\Phi(z)$	0.6915	0.8413	0.8997	0.9332	0.9500	0.9750	0.9772	0.9938	0.9951	0.9987

Use $P(Z > z) = 1 - \Phi(z)$ and, by symmetry, $\Phi(-z) = 1 - \Phi(z)$.

- **Question mix:** Q1 derivation (OLS FOCs and residual identities); Q2 regression-output interpretation (prediction, change in prediction, units); Q3 large-sample sampling distribution of $\hat{\beta}_1$ (standardising, probability calculations, the role of $SE(\hat{\beta}_1)$); Q4 binary regressor (group means, an OLS-with-a-dummy proof, Least Squares Assumption 1 in this setting); Q5 conceptual (causality, Least Squares Assumption 1, selection bias, Type I/II trade-off).

Instructions

Answer **all** questions. Show all working clearly; answers without justification may receive zero marks.

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Question 1**(10 Marks)**

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, \dots, n.$$

For candidate values b_0, b_1 , define the sum of squared errors

$$S(b_0, b_1) = \sum_{i=1}^n (Y_i - b_0 - b_1 X_i)^2.$$

The OLS estimators $\hat{\beta}_0, \hat{\beta}_1$ are the values of b_0, b_1 that minimize $S(b_0, b_1)$.

- (a) (4 marks) Derive the two first-order conditions for this minimization problem, and show that they can be written as

$$\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.$$

- (b) (3 marks) Using part (a), show that the OLS residuals $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ satisfy

$$\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0.$$

- (c) (3 marks) Using part (b), show that $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$, where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ is the OLS predicted value.

Solution**Method 1: Direct approach**

(a)

$$\begin{aligned} \left. \frac{\partial S}{\partial b_0} \right|_{(\hat{\beta}_0, \hat{\beta}_1)} &= -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0 \\ \left. \frac{\partial S}{\partial b_1} \right|_{(\hat{\beta}_0, \hat{\beta}_1)} &= -2 \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0 \end{aligned}$$

$$\div (-2) : \quad \boxed{\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \quad \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) X_i = 0.}$$

(b)

$$\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i \text{ in (a)} \implies \boxed{\sum_{i=1}^n \hat{u}_i = 0, \quad \sum_{i=1}^n \hat{u}_i X_i = 0.}$$

(c)

$$\sum_{i=1}^n \hat{u}_i \hat{Y}_i = \hat{\beta}_0 \sum_{i=1}^n \hat{u}_i + \hat{\beta}_1 \sum_{i=1}^n \hat{u}_i X_i = \boxed{0} \quad (\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i; \text{ (b)})$$

Marking scheme (10 marks)**(a) (4 marks)** First-order conditions

- 2 marks: correct partial derivative of S with respect to b_0 , set to zero.
- 2 marks: correct partial derivative of S with respect to b_1 , set to zero, and both simplified (divided by -2) to the stated form.

(b) (3 marks) Residual identities

- 1 mark: correctly identifies $\hat{u}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ as the object appearing in the FOCs.
- 2 marks: correctly substitutes to obtain both $\sum_{i=1}^n \hat{u}_i = 0$ and $\sum_{i=1}^n \hat{u}_i X_i = 0$ (1 mark each).

(c) (3 marks) $\sum \hat{u}_i \hat{Y}_i = 0$

- 1 mark: correctly expands $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ and splits the sum.
- 1 mark: correctly invokes both identities from part (b).
- 1 mark: correct final conclusion $\sum_{i=1}^n \hat{u}_i \hat{Y}_i = 0$.

Question 2**(10 Marks)**

A researcher studying the Singapore private rental market collects data on $n = 150$ HDB flat rental listings and regresses monthly rent (in SGD) on floor area (in square metres, sqm). The estimated OLS regression line is

$$\widehat{\text{Rent}}_i = 380 + 24.5 \text{Area}_i.$$

- (a) (3 marks) A 90 sqm flat is listed for rent. What is the regression's predicted monthly rent for this flat?
- (b) (3 marks) A tenant moves from a 70 sqm flat to a 95 sqm flat. What is the regression's predicted change in monthly rent?
- (c) (4 marks) Interpret the slope coefficient, including units. Is the intercept economically meaningful in this context? Explain briefly.

Solution**Method 1: Direct approach**

(a)

$$380 + 24.5(90) = 2585 \implies \boxed{\widehat{\text{Rent}} = \text{SGD } 2,585 \text{ per month}}$$

(b)

$$\hat{\beta}_1 \Delta \text{Area} = 24.5 \times (95 - 70) = 24.5 \times 25 = 612.5$$

$$\implies \boxed{\Delta \widehat{\text{Rent}} = \text{SGD } 612.50 \text{ (an increase)}}$$

- (c)
 - Slope: +SGD 24.50 per month in predicted rent per extra sqm of floor area.
 - Intercept: prediction at Area = 0, which never occurs in the data $\implies$ no economic meaning.

Slope: +SGD 24.50/month per extra sqm; intercept not economically meaningful.

Marking scheme (10 marks)(a) **(3 marks)** Predicted rent

- 1 mark: correct substitution Area = 90 into the regression line.
- 1 mark: correct arithmetic.
- 1 mark: final answer stated with correct units (SGD/month).

(b) **(3 marks)** Predicted change

- 1 mark: correct method (slope $\times$ change in Area, or difference of two predictions).
- 1 mark: correct arithmetic ($\Delta \text{Area} = 25$, product = 612.5).
- 1 mark: correctly stated with units and direction (increase).

(c) **(4 marks)** Interpretation

- 2 marks: correct slope interpretation with units and correct sign/direction.
- 2 marks: correct discussion that the intercept is not economically meaningful (Area = 0 does not occur in the data), with brief justification.

Question 3

(10 Marks)

A team of researchers studies $n = 250$ Singapore tech start-ups, regressing monthly revenue growth (%) on monthly R&D spending (SGD '000s). Based on the large sample, they treat the OLS estimator $\hat{\beta}_1$ as following its large-sample sampling distribution

$$\hat{\beta}_1 \sim N(\beta_1, \sigma_{\hat{\beta}_1}^2),$$

with population slope $\beta_1 = 2.0$ and standard deviation of the sampling distribution $\sigma_{\hat{\beta}_1} = \text{SE}(\hat{\beta}_1) = 0.5$.

- (2 marks) Explain, in words, why $\hat{\beta}_1$ – computed from this one sample of 250 start-ups – is itself a random variable, even though the population slope β_1 is a fixed (if unknown) number.
- (3 marks) Using the large-sample result $\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\hat{\beta}_1}} \sim N(0, 1)$, standardise and use the standard normal table in the Overview to compute $P(\hat{\beta}_1 > 2.75)$.
- (3 marks) Compute $P(|\hat{\beta}_1 - \beta_1| < 0.98)$, i.e. the probability that the estimator falls within 0.98 of the true population slope. Use the standard normal table in the Overview.
- (2 marks) Explain what $\text{SE}(\hat{\beta}_1)$ measures, and give two reasons – referring to the formula for $\sigma_{\hat{\beta}_1}^2$ – why it becomes *smaller* as (i) the sample size n increases, and (ii) the sample spread of the regressor X increases.

Solution

Method 1: Direct approach

- A new sample gives a new $\hat{\beta}_1$ (a function of the random sample); β_1 is fixed.

$\hat{\beta}_1$ is a function of the random sample $(Y_i, X_i)_{i=1}^n$, so it varies across samples and is itself random.

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$$P(\hat{\beta}_1 > 2.75) = P\left(\frac{\hat{\beta}_1 - 2.0}{0.5} > \frac{2.75 - 2.0}{0.5}\right) = P(Z > 1.5) \quad (Z \sim N(0, 1))$$

$$= 1 - \Phi(1.5) = 1 - 0.9332 \implies \boxed{P(\hat{\beta}_1 > 2.75) = 0.0668}$$

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$$\begin{aligned} P(|\hat{\beta}_1 - 2.0| < 0.98) &= P\left(\left|\frac{\hat{\beta}_1 - 2.0}{0.5}\right| < \frac{0.98}{0.5}\right) = P(|Z| < 1.96) \\ &= 2\Phi(1.96) - 1 = 2(0.9750) - 1 \quad (\text{symmetry}) \\ &\implies \boxed{P(|\hat{\beta}_1 - \beta_1| < 0.98) = 0.9500} \end{aligned}$$

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$$\sigma_{\hat{\beta}_1}^2 = \frac{1}{n} \cdot \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 u_i^2}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^2}$$

- $SE(\hat{\beta}_1)$ estimates $\sigma_{\hat{\beta}_1}$, the spread of $\hat{\beta}_1$ across samples (its precision).
- (i) Larger n : the factor $1/n$ shrinks $\sigma_{\hat{\beta}_1}^2$.
- (ii) More spread in X : with $s_X^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$, the numerator is $\propto s_X^2$ and the denominator is s_X^4 , so $\sigma_{\hat{\beta}_1}^2 \propto 1/s_X^2$ falls.

$SE(\hat{\beta}_1)$ measures the spread of $\hat{\beta}_1$'s sampling distribution;
it falls as n rises and as the spread of X rises.

Marking scheme (10 marks)

(a) **(2 marks)** Why $\hat{\beta}_1$ is random

- 1 mark: recognizes $\hat{\beta}_1$ is a function of the random sample (Y_i, X_i) .
- 1 mark: correctly explains that a different sample gives a different numerical value, hence a sampling distribution, while β_1 stays fixed.

(b) **(3 marks)** $P(\hat{\beta}_1 > 2.75)$

- 1 mark: correct standardisation, $z = 1.5$.
- 1 mark: correctly reads $\Phi(1.5) = 0.9332$ from the table.
- 1 mark: correct final probability 0.0668.

(c) **(3 marks)** $P(|\hat{\beta}_1 - \beta_1| < 0.98)$

- 1 mark: correct standardisation to $P(|Z| < 1.96)$.
- 1 mark: correct use of symmetry, $2\Phi(1.96) - 1$.
- 1 mark: correct final probability 0.9500.

(d) **(2 marks)** Meaning of $SE(\hat{\beta}_1)$ and why it shrinks

- 1 mark: correctly states $SE(\hat{\beta}_1)$ measures the spread/dispersion of the sampling distribution of $\hat{\beta}_1$ (precision of the estimator).
- 1 mark: valid explanation of *both* why it falls with larger n and with more spread in X .

Question 4

(10 Marks)

A study of $n = 200$ food-delivery riders in Singapore regresses monthly income (in SGD) on D_i , a dummy variable equal to 1 if the rider owns their vehicle and 0 if the rider rents it ($n_0 = 120$ renters, $n_1 = 80$ owners). The estimated regression is

$$\widehat{\text{Income}}_i = 1450 + 320 D_i, \quad \text{SE}(\hat{\beta}_1) = 95.$$

- (3 marks) State the average monthly income predicted by the regression for renters and for owners.
- (2 marks) Give an economic interpretation of $\hat{\beta}_1 = 320$.
- (3 marks) Prove, using the OLS normal equations from Question 1 applied to this binary regressor $D_i \in \{0, 1\}$, that in general $\hat{\beta}_0 = \bar{Y}_0$ (the sample mean of income among renters) and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$ (the difference in sample mean income between owners and renters), where $\bar{Y}_0, \bar{Y}_1$ are the group sample means and n_0, n_1 are the group sizes.
- (2 marks) State what Least Squares Assumption 1 ($E[u_i | D_i] = 0$) requires in this dummy-regressor setting, give one specific unobserved factor that could plausibly make it fail here, and state the likely direction of the resulting bias in $\hat{\beta}_1$.

Solution

Method 1: Direct approach

- $D = 0$: $1450 + 320(0) = 1450$; $D = 1$: $1450 + 320(1) = 1770$.

Renters: SGD 1,450/month; Owners: SGD 1,770/month.

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$$\hat{\beta}_1 = 1770 - 1450 = 320 \implies \text{Riders who own their vehicle earn, on average, SGD 320/month more than those who rent.}$$

- Normal equations (Q1) with $D_i^2 = D_i$, $\sum_{i=1}^n D_i = n_1$, $\sum_{i=1}^n Y_i D_i = n_1 \bar{Y}_1$, $n = n_0 + n_1$:

$$n_0 \bar{Y}_0 + n_1 \bar{Y}_1 = n \hat{\beta}_0 + n_1 \hat{\beta}_1 \quad (\text{NE1; } \sum_{i=1}^n Y_i \text{ split by group})$$

$$n_1 \bar{Y}_1 = n_1 \hat{\beta}_0 + n_1 \hat{\beta}_1 \implies \bar{Y}_1 = \hat{\beta}_0 + \hat{\beta}_1 \quad (\text{NE2; } n_1 > 0)$$

$$n_0 \bar{Y}_0 = n_0 \hat{\beta}_0 \quad (\text{NE1-NE2})$$

$$\implies \hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_1 = \bar{Y}_1 - \hat{\beta}_0 = \bar{Y}_1 - \bar{Y}_0 \quad (n_0 > 0)$$

$$\hat{\beta}_0 = \bar{Y}_0 \text{ (renter mean),} \quad \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0 \text{ (owner mean - renter mean).}$$

- Least Squares Assumption 1 requires $E[u_i | D_i = 0] = 0$ and $E[u_i | D_i = 1] = 0$: no average difference between owners and renters in unobserved determinants of income.
 - Fails if riders who choose to own are more experienced/committed: $E[u_i | D_i = 1] > E[u_i | D_i = 0]$.

- That factor is positively correlated with D_i and income $\implies \hat{\beta}_1$ biased upward.

$$E[u \mid D] \neq 0 \text{ (self-selection into ownership by experience/commitment)} \\ \implies \hat{\beta}_1 \text{ is upward biased.}$$

Method 2: group-wise minimisation of S (part (c))

$$c_0 := b_0, \quad c_1 := b_0 + b_1 \quad (\text{bijection of } \mathbb{R}^2)$$

$$S = \sum_{D_i=0} (Y_i - c_0)^2 + \sum_{D_i=1} (Y_i - c_1)^2$$

$$\implies \text{each sum minimised at its group mean } (\sum (Y_i - c)^2 \text{ min at } \bar{Y}; n_0, n_1 > 0) \\ \implies \hat{\beta}_0 = \bar{Y}_0, \quad \hat{\beta}_0 + \hat{\beta}_1 = \bar{Y}_1 \implies \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$$

Marking scheme (10 marks)

(a) (3 marks) Group means

- 1 mark: correct method ($D = 0$ and $D = 1$ substituted into the regression line).
- 1 mark: correct renter mean (1450).
- 1 mark: correct owner mean (1770).

(b) (2 marks) Interpretation of $\hat{\beta}_1$

- 1 mark: recognizes $\hat{\beta}_1$ as the difference in group means.
- 1 mark: correct direction/magnitude with units (owners earn SGD 320 more).

(c) (3 marks) Proof that $\hat{\beta}_0 = \bar{Y}_0, \hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$

- 1 mark: correctly uses the first normal equation (splitting $\sum_{i=1}^n Y_i$ by group and using $\sum_{i=1}^n D_i = n_1$).
- 1 mark: correctly uses the second normal equation, including $D_i^2 = D_i$, to obtain $\bar{Y}_1 = \hat{\beta}_0 + \hat{\beta}_1$.
- 1 mark: correct algebra combining the two to reach $\hat{\beta}_0 = \bar{Y}_0$ and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$.

(d) (2 marks) Least Squares Assumption 1

- 1 mark: correctly states Assumption 1 as requiring $E[u_i \mid D_i = 0] = 0$ and $E[u_i \mid D_i = 1] = 0$ separately.
- 1 mark: valid confounding factor with correct direction of bias.

Question 5

(10 Marks)

- (a) (4 marks) A researcher regresses PSLE (Primary School Leaving Examination) score on weekly hours of private tuition, using observational data on Singapore primary-six pupils. She finds a strong positive coefficient on tuition hours and concludes that private tuition *causes* higher PSLE scores. Using Least Squares Assumption 1 ($E[u_i|X_i] = 0$) and the concept of selection bias, explain why this conclusion may not be valid. In your answer, state a specific unobserved factor that could be correlated with tuition hours, and predict the likely direction of the resulting bias.
- (b) (3 marks) Suggest one research design that would let the researcher obtain a more credible estimate of the causal effect of tuition on PSLE score, and briefly explain why it addresses the problem identified in part (a).
- (c) (3 marks) A bank's automated loan-approval system flags each application as "high risk" (reject) or "safe" (approve) based on a credit-score cutoff. Define, in the context of this bank's decision, what a Type I error and a Type II error would mean, and explain the trade-off the bank faces if it raises the credit-score cutoff required for approval.

Solution

Method 1: Direct approach

- (a)
- Least Squares Assumption 1: $E[u_i | \text{Tuition}_i] = 0$ in $\text{PSLE}_i = \beta_0 + \beta_1 \text{Tuition}_i + u_i$.
 - Parental income/motivation (or ability) sits in u_i , is correlated with tuition hours, and raises PSLE regardless: selection bias, so $E[u_i | \text{Tuition}_i]$ rises with tuition and Assumption 1 fails.
 - The factor is positively correlated with both tuition and PSLE $\implies \hat{\beta}_1$ biased upward.

$$\begin{aligned} E[u | \text{Tuition}] &\neq 0 \text{ (selection on family resources and ability)} \\ &\implies \hat{\beta}_1 \text{ is upward biased.} \end{aligned}$$

- (b) RCT: free tuition slots randomly assigned to eligible pupils.

$$\text{Tuition}_i \perp u_i \text{ (family income, motivation, ability)}$$

$$\implies E[u_i | \text{Tuition}_i] = E[u_i] = 0 \text{ (Assumption 1 restored)}$$

$$\begin{aligned} \text{Random assignment of tuition} &\implies E[u | \text{Tuition}] = 0 \\ &\implies \text{causal interpretation valid.} \end{aligned}$$

- (c)
- Type I: a truly safe applicant is flagged "high risk" and rejected (lost interest income).
 - Type II: a truly high-risk applicant is flagged "safe" and approved (later default).

$$\begin{aligned} \text{Higher cutoff: } &\downarrow \text{ Type II error (fewer bad loans),} \\ &\uparrow \text{ Type I error (more good applicants rejected).} \end{aligned}$$

Marking scheme (10 marks)**(a) (4 marks) Selection bias**

- 1 mark: correctly states Assumption 1 ($E[u | X] = 0$) as the relevant condition.
- 1 mark: identifies a specific, plausible confounding factor (e.g. family income, motivation, or ability).
- 1 mark: explains why the confounder is correlated with tuition hours (self-selection).
- 1 mark: correct direction of bias (upward) with valid justification.

(b) (3 marks) Alternative design

- 1 mark: proposes a valid design (RCT, natural experiment, or instrumental variable).
- 1 mark: explains that random/quasi-random assignment breaks the correlation between tuition and u .
- 1 mark: correctly links this back to Assumption 1 being restored / causal interpretation being valid.

(c) (3 marks) Type I/II trade-off

- 1 mark: correct Type I error definition in context.
- 1 mark: correct Type II error definition in context.
- 1 mark: correct description of the trade-off when the cutoff is raised.